

## PROBABILITY

**Probability** is a branch of mathematics that deals with the likelihood or chance of an event occurring. It quantifies uncertainty and is a fundamental concept in statistics, data analysis, and decision-making.

### Important concepts

1. **Experiment:** An action or process with uncertain outcomes, such as rolling a die or tossing a coin, selecting a card from deck of cards.
2. **Sample Space (S):** The set of all possible outcomes of an experiment.
  - Example: For rolling a die,  $S = \{1, 2, 3, 4, 5, 6\}$
3. **Event:** A subset of the sample space that includes a specific outcome.  
Example: When a dice is rolled,  $P(\text{Getting an even number}) = 3/6$

### Probability of an Event P(E):

A measure of how likely an event is to occur. It is a number between 0 and 1, where:

$P(E) = 0$ : The event is impossible.

$P(E) = 1$ : The event is sure.

$P(E)$  : The closer to 1, the more likely the event is to occur

$$\text{Probability(Event)} = \frac{\text{no.of favourable outcomes}}{\text{total no.of outcomes}}, \quad P(E) = \frac{n(E)}{n(S)} \quad \text{where } 0 \leq P(E) \leq 1$$

- **Complementary Events:** For event E,  $E^c$  is the complementary event such that  $P(E) + P(E^c) = 1$
- **Mutually Exclusive events:** Two events A and B are said to be mutually exclusive if they cannot occur at the same time. If one occurs, the other is impossible.  
**Ex:** When a coin is tossed, head or tail may not occur at the same time.
- **Independent Events:** Two events where one does not affect the other. Both can occur but occurrence of one gives no information about other. Ex. A coin is tossed to get H/T cannot affect the outcome in next toss
- **Equally Likely Events:** A given number of events are said to be equally likely if none of them is expected to occur in preference to the others.  
**Ex:** if a unbiased dice is rolled, each outcome is equally likely to occur.

#### **Mutually exclusive events**

$P(A \cap B) = 0$ , they cannot occur together

$$P(A \cup B) = P(A) + P(B)$$

#### **Independent events**

$P(A \cap B) = P(A) \times P(B)$ , multiply directly

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$P(A/B) = 0$ , if B happened A is impossible

$P(A/B) = P(A)$ , B gives no info for A

### Types of Probability:

- **Theoretical Probability:** Based on reasoning or a mathematical model. Example:  
Probability of rolling a 3 on a die =  $\frac{1}{6}$
- **Experimental (Empirical) Probability:** Based on the relative frequency of the event occurring during repeated trials. Example: If you roll a die 100 times and get a 3 in 20 rolls,  $P(3) = \frac{20}{100} = 0.2$
- **Subjective Probability:** Based on personal judgment or experience, without precise calculations.

Probabilities may be either marginal, joint or conditional. Understanding their differences and how to manipulate among them is key to success in understanding the foundations of statistics.

**Marginal probability:** refers to finding the probability of one event in isolation, regardless of other event or probability of single event occurring.

Example:  $P(A)$  is the marginal probability for event A. It is an unconditional probability, not conditioned on another event.

probability that a card drawn is red ( $P(\text{red}) = 0.5$ ).

probability that a card drawn is a 4 ( $P(\text{four}) = 1/13$ ).

**Joint probability:** is the probability of two (or more) events occurring simultaneously. It is the probability of the intersection of two or more events. If A and B are two events, their joint probability is  $P(A \cap B)$  means P(A and B)

Example: the probability that a card is a four and red

$P(\text{four and red}) = 2/52 = 1/26$ . (There are two red fours in a deck of 52, the 4 of hearts and the 4 of diamonds).

**Conditional probability:** It is the probability of event A occurring, given that event B has already occurred. This is expressed as  $P(A|B)$  ("the probability of A given B").

$$P(A|B) = P(A \cap B) / P(B)$$

Read as: "Probability of A given B"

Example: Given that you drew a red card, what's the probability that it's a four  
 $P(\text{four} | \text{red}) = 2/26 = 1/13$ . So out of the 26 already chosen red cards, there are two fours so  
 $2/26 = 1/13$ .

### **Multiplication Theorem on conditional probability**

For events A and B,

$$P(A \cap B) = P(A) \cdot P(B/A) = P(B) \cdot P(A/B)$$

For events A, B, C

$$P(A \cap B \cap C) = P(A) \cdot P(B/A) \cdot P(C/(A \cap B))$$

## **Probability Rules**

### **1. Addition Rule**

The addition rule is used when probability of at least one of the two events is required.

For mutually exclusive events,  $P(A \cap B) = 0$ , so  $P(A \cup B) = P(A) + P(B)$

For non - mutually exclusive events:  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Q What is the probability that the selected card is a king or a diamond.

$$P(\text{king}) = 4/52; P(\text{diamond}) = 13/52; P(\text{diamond king}) = 1/52$$

$$P(K \cup D) = P(K) + P(D) - P(K \cap D) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13$$

### **2. Multiplication Rule**

The multiplication rule is used when the probability of both events happened together is required.

$$\text{for independent events: } P(A \cap B) = P(A) \cdot P(B)$$

$$\text{For dependent events: } P(A \cap B) = P(A) \cdot P(B|A)$$

Q What is the probability of getting A grade I both maths and science if  $P(\text{getting in Math}) = 0.7$  and  $P(\text{getting in Science}) = 0.6$ .

$$P(M) \times P(S) = 0.7 \times 0.6 = 0.42$$

### 3. Odds in favor / against

For an event  $E$ :

Probability compares “favourable outcomes” to all outcomes.

Odds compare “favourable outcomes” to “unfavourable outcomes”.

**So odds are a ratio of success: failure, not success: total.**

Odds of an event is the ratio of the number of observations in favor of event to the total number of observations not in favor of the event.

**Odds in favor of event A are a:b**

**Odds against event A are b:a where  $P(A) = a/(a+b)$**

Q The odds that a person picked at random will be either overweight or obese are 14:11. What is the probability that the person is fit (is not overweight or obese)

Odds in favor of A are 14:11, so  $P(A) = 14/(14 + 11) = 14/25$  and  $P(A') = 1 - 14/25 = 11/25$

### 4. Conditional Probability:

The probability of event B given that A has occurred:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad (\text{Probability of B when A has already occurred})$$

Q What is the probability of getting a king given that it is a card of hearts.

$$P(K) = 4/52; P(H) = 13/52; P(K \cap H) = 1/52,$$

$$\text{so } P(K|H) = P(K \cap H) / P(H) = (1/52) / (13/52) = 1/13$$

## BAYES'S THEOREM

**Bayes' Theorem describes the relationship between joint, marginal, and conditional probabilities.**

It is used to update the probability of an event (or hypothesis) based on new evidence.

**It allows for the "inversion" of conditional probabilities, meaning you can calculate  $P(A|B)$  if you know  $P(B|A)$**

Bayes' theorem is an equation that allows us to manipulate conditional probabilities.

For two events A and B, Bayes' theorem lets us to go from  $P(B|A)$  to  $P(A|B)$ , if we know the marginal probabilities of the outcomes of A,  $P(A)$  and the conditional probability of B, given the outcomes of A,  $P(B|A)$

Here is the equation for Bayes' theorem for two events with two possible outcome (A and not A).

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B|A)}{P(B)}$$

**where:**

$P(A)$  = The probability of A occurring

$P(B)$  = The probability of B occurring

$P(A|B)$  = The probability of A given B

$P(B|A)$  = The probability of B given A

$P(A \cap B)$  = The probability of both A and B occurring

## PERMUTATION AND COMBINATION IN BAYES THEOREM

**Permutation is arrangement of items where order matters**

$$P(n, r) = n! / (n-r)! \quad {}^n P_r = \frac{n!}{(n-r)!}$$

**Combination is selection of items where order does not matter.**

$$C(n, r) = n! / r!(n-r)! \quad {}^n C_r = \frac{n!}{r!(n-r)!}$$

Permutations and Combinations are used in Bayes' Theorem to **calculate the individual probability terms** — especially  $P(B|A)$ ,  $P(A)$ , and  $P(B)$  — when the sample space involves counting arrangements or selections.

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| Formula                                   | Name               | Use When                       |
|-------------------------------------------|--------------------|--------------------------------|
| $P(A) = f/n$                              | Basic              | Simple probability             |
| $P(A') = 1 - P(A)$                        | Complement         | Finding opposite event         |
| $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ | Addition           | Events can overlap             |
| $P(A \cup B) = P(A) + P(B)$               | Mutually Exclusive | Events cannot overlap          |
| $P(A \cap B) = P(A) \times P(B)$          | Independent        | Events don't affect each other |
| $P(A \cap B) = P(A) \times P(B A)$        | Dependent          | Events affect each other       |
| $P(A B) = P(A \cap B) / P(B)$             | Conditional        | One event given another        |
| $P(A B) = P(B A) \times P(A) / P(B)$      | Bayes              | Update belief with evidence    |
| $nPr = n! / (n-r)!$                       | Permutation        | Ordered arrangements           |
| $nCr = n! / r!(n-r)!$                     | Combination        | Unordered selections           |
| $P(X=r) = C(n,r)p^r(1-p)^{n-r}$           | Binomial           | Fixed trials, two outcomes     |
| $P(X=k) = e^{-\lambda} \lambda^k / k!$    | Poisson            | Rate-based events              |

**Ques:** A bag contains 5 red and 3 blue cards. Two cards are drawn at random without replacement. Find the probability that first card is red when both cards are red.

Let A be the event of drawing first red card.

Let B be the event of drawing second red card.

$$P(A) = C(5,1) / C(8,1) = 5/8 = 0.625$$

$$P(B/A) = C(4,1) / C(7,1) = 4/7 = 0.571$$

Case I

Ist red card, IInd also red

$$5/8 \times 4/7 = 20/56$$

$$\text{So } P(B) = 20/56 + 15/56 = 35/56 = 0.625$$

Applying Bayes theorem

$$P(A/B) = [P(B/A) \times P(A)] / P(B)$$

Case II

Ist blue card, IInd red card

$$3/8 \times 5/7 = 15/56$$

$$(20/56) / (35/56) = 20/35 = 4/7 = 0.571$$

So probability that first card is red is 0.571

### **MORE EXAMPLES:**

Q1. 20 cards are numbered from 1 to 20. One card is drawn at random. What is the probability that the number on card drawn is:

- a) A prime no.    b) an odd number    c) a multiple of 5    d) not divisible by 3

Q2. A coin is tossed two times. What is the probability of getting:

- 1) Two heads    2) one tail    3) No head    4) At least one head    5) At most one head

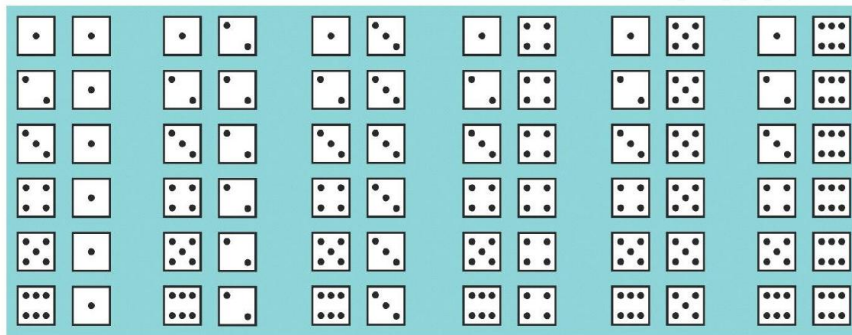
Q3 A card is drawn from a pack of 52 playing cards. What is the probability of getting....

- 1) red colour card    b) ace of black    3) a king or a queen    4) jack of diamond
- 5) a spade    6) a black card and a queen

Q4. Two dice are thrown simultaneously. Find the probability of getting :

- a) a doublet    b) an even no. as sum    c) a multiple of 3 as sum    d) a total of atleast 10
- e) a doublet of even no.    f) a multiple of 2 on one dice and multiple of 3 on other.

Q5 In a single throw of two dice, find the probability that neither a doublet nor a total of 10 will appear.



### **For conditional probability**

Q Suppose the probability that event A occurs is 0.5 and the probability that event B occurs is 0.3. If A and B are mutually exclusive, what is the probability that neither A nor B occurs? [0.2]

Q4 A die is rolled. If the outcome is an odd number, what is the probability that it is prime. (2/3)

Q5 Assume that each born child is equally likely to be a boy or a girl. If a family has two children, what is the conditional probability that both are girls given that (i) the youngest child is girl, (ii) at least one of the children is a girl?  $[1/2, 1/3]$

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Q6 Ramesh appears for an interview for two posts A and B, for which the selection is independent. The probability for his selection for post A is  $(1/6)$  and for B is  $(1/7)$ . Find the probability that Ramesh is selected for at least one post.  $(2/7)$

Q7 A problem is given to three students whose chance of solving it are  $1/3, 2/7, 3/8$ . What is the probability that the problem will be solved.  $(59/84)$

### **THEOREM OF TOTAL PROBABILITY**

Let  $E_1, E_2, E_3, \dots$  be mutually exclusive events associated with a random experiment and let E be the event that occurs with some  $E_i$  then,

$$P(E) = \sum P(E/E_i) \cdot P(E_i)$$

Q8: There are 3 urns containing 3 white and 2 black balls; 2 white and 3 black balls; 1 black and 4 white balls respectively. There is equal probability of each urn being chosen. One ball is drawn from an urn chosen at random. What is the probability that a white ball is drawn.  $(3/5)$

Q9 A factory has 3 machines, X, Y, Z producing 1000, 2000, 3000 bolts per day respectively. The machine X produces 1% defective bolts, Y produces 1.5% defective bolts and Z produces 2% defective bolts. At the end of the day, a bolt is drawn at random and it is found to be defective. What is the probability that this defective bolt has been produced by machine A.  $(1/10)$

Q10 An insurance company insured 2000 scooter drivers, 4000 car drivers and 6000 truck drivers. The probability of an accident involving a scooter, a car and a truck is  $1/100, 3/100, 3/20$  respectively. One of the insured persons meet with an accident. What is the probability that he is a scooter driver?  $(1/52)$